Parameter-Efficient Reparameterization Tuning for Remote Sensing Image–Text Retrieval

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***Abstract*— Vision language models have been gradually adapted to various tasks of remote sensing (RS) domain with full fine-tuning paradigm, e.g., RS image–text retrieval (RSITR). Superior performance enhancements have proven the powerful generalization and robustness of vision language models. However, full fine-tuning vision language models is resource-intensive and poses risks of overfitting. Moreover, existing RSITR methods usually assume that RS images correspond to text captions one by one and utilize the bidirectional matching (BiMa) training objective, which is not aligned with evaluation benchmarks and real-world applications. To tackle the mentioned problems, we propose a novel parameter-efficient reparameterization tuning with ranking and matching (PERT-RaMa) framework, which effectively migrates the vision language model [i.e., contrastive language-image pre-training (CLIP)] to RSITR task. To overcome the overfitting issue, we build a lightweight, plug-and-play module called Kronecker product for low-rank adaptation (KPLoRA). KPLoRA obtains a higher intrinsic rank with fewer parameters. Furthermore, we design the RaMa training method that converts RSITR task into one-to-one matching and one-to-many ranking, which is aligned with current RSITR benchmarks and accelerates training speed through removal of unessential computations. Comprehensive experiments on three public RSITR benchmarks demonstrate that the effectiveness and efficiency of the proposed retrieval model. Our method outperforms full fine-tuning methods without CLIP by nearly 5%–10%, and achieves comparable or superior retrieval capability than CLIP with full fine-tuning and parameter-efficient fine-tuning (PEFT). Furthermore, our RaMa training method increases the training speed by 5× compared to current training method.**

***Index Terms*— Kronecker product, low-rank adaptation (LoRA), parameter-efficient reparameterization tuning (PERT), ranking and matching (RaMa), remote sensing image–text retrieval (RSITR).**

I. Introduction

ADVANCEMENTS in Earth observation technology have significantly increased the accessibility of remote sensing (RS) imagery over the past decades [1]. The massive RS images have facilitated potential applications in environmental monitoring, urban planning, and smart agriculture [2], [3], [4], [5]. However, a large amount of redundant data interferes with the extraction and utilization of useful information. How to automatically select meaningful and valuable RS images from large RS archives has becomes a critical challenges and hot topic.

RS image–text retrieval (RSITR) aims to search and retrieve corresponding RS images with text queries. RSITR establishes effective relation between RS images and human-understandable natural language, providing a powerful tool for understanding the Earth and mining useful information [6]. Compared with traditional content-based RS image retrieval [7], [8], RSITR is more efficient and user-friendly due to the semantic richness and practicality in human–computer interaction of text descriptions.

Previous RSITR methods [9], [10], [11] typically employ separate visual and textual encoders to extract features, and then map the features into a joint feature space. Some methods also introduce cross-modal interaction modules [9], [12] to bridge the semantic gap between image and text modality. These prior methods adopt the full fine-tuning paradigm, which optimizes all parameters of the retrieval model during the training process. However, due to the limited scale and representation capabilities of widely used feature extraction models (e.g., ResNet [13] and GRU [14]), the existing RSITR methods still fail to achieve satisfactory retrieval performance despite the resource-intensive full fine-tuning paradigm.

As the scale of image–text pairs expands, vision language models emerge with superiority and generalization in visual tasks [15], [16] and cross-modal image–text tasks [17], [18]. Concretely, contrastive language-image pre-training (CLIP) [19] is trained on 400 million natural image–text pairs for bidirectional matching (BiMa), demonstrating considerable potential for task adaptation [20] and domain adaptation [21]. CLIP with full fine-tuning has shown superior performance in both natural domain [22] and RS domain,[1], while the computational cost and memory usage make it impractical in real-world RS applications. Parameter-efficient fine-tuning (PEFT) paradigm offers a resource-friendly alternative for the domain adaptation of CLIP.

PEFT optimizes a small set of inherent or extra parameters and freezes the rest of the parameters. Currently, PEFT is

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[1]https://github.com/arampacha/CLIP-rsicd

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divided into addition tuning, specification tuning, and reparameterization tuning [23]. PE-RSITR [6] has explored the additional tuning for RSITR, introducing an extra cross-modal adapter into CLIP. PE-RSITR fully demonstrates that the PEFT paradigm for CLIP can effectively bridge domain differences between natural and RS domain, but it increases model complexity and inference latency due to additional modules. Parameter-efficient reparameterization tuning (PERT) equivalently fine-tunes the model through the purely linear transformation with a low dimension, which does not introduce any additional layers or modules. Low-rank adaptation (LoRA) [24], as the most representative and widely used PERT form, has not yet been explored in RSITR. However, existing PERT methods [24], [25], [26], [27] mainly focus on downstream tasks of natural language processing. Therefore, some challenges remain to be addressed when exploring PERT and RSITR.

LoRA optimizes a feature subspace by leveraging a low intrinsic rank, yet the transfer effectiveness of LoRA-style PERT methods for specific downstream tasks (e.g., RSITR) strongly depends on the chosen intrinsic rank. It only captures limited information and yields suboptimal performance with low intrinsic rank, while the high intrinsic rank may introduce redundant parameters and increase the risk of overfitting. To address the difficulty, we design a LoRA-style module for RSITR that achieves a higher intrinsic rank with fewer parameters, thereby mitigating the overfitting issue and enhancing transfer performance.

Meanwhile, we revisit the task modeling and training approach of RSITR. Existing RSITR methods [6], [9], [11] typically optimize retrieval models using a symmetric BiMa objective, assuming the task as bidirectional one-to-one image–text matching. However, it is misaligned with standard RSITR evaluation benchmarks [9], [28], [29], where each RS image is paired with multiple captions (often five), rather than one-to-one image–text pairs. As a result, the current BiMa training approach redundantly pairs each RS image with multiple text captions within an epoch, leading to repetitive computations of RS image feature extraction. Motivated by this inefficiency, this article remodels the task and designs a novel training approach tailored to current RSITR benchmarks, aiming to reduce redundancy and enhance model efficiency.

To alleviate the above problems, this article proposes a novel PEFT framework, named PERT with ranking and matching (PERT-RaMa). First, we build a novel algorithmic framework for PEFT, which supports various LoRA-style PERT modules in a plug-and-play manner. Then, we propose the Kronecker product for LoRA (KPLoRA) module. KPLoRA utilizes the Kronecker product to reorganize the LoRA module, leveraging its ability of rank expansion to learn the feature subspace of higher intrinsic rank with fewer trainable parameters. KPLoRA further employs the cross-layer sharing mechanism, which enhances the retrieval performance for RSITR. Moreover, this article transforms the RSITR task into a RaMa problem and designs an efficient training method, named RaMa. RaMa effectively boosts the training speed and recall with a consistent asymmetric loss function.

The main contributions of this article can be summarized as follows.

1) We propose a novel PERT-RaMa framework for RSITR task in the parameter-efficient form. The proposed framework is composed of the dual-branch CLIP encoders, KPLoRA, and the RaMa training method.
2) We develop a LoRA-style module for RSITR, KPLoRA. We employ the Kronecker product to reconstruct the LoRA module. We analyze and compare KPLoRA and several representative LoRA-style modules from the perspective of intrinsic rank. Quantitative results demonstrate that KPLoRA achieves better retrieval recall and overcomes the overfitting problem with various intrinsic ranks.
3) We review the current task modeling and training method of RSITR. We designed an effective and efficient training method, named RaMa. Ablation experiments prove that RaMa significantly outperforms the current training method in training speed and retrieval capability.
4) We provide a broad range of empirical studies for specification tuning and reparameterization tuning methods on RSITR. Extensive experiments have advanced researches on PEFT for RSITR tasks.

The rest of the article is organized as follows. First, Section II reviews the relevant researches on RSITR and PEFT. Then, Section III presents the proposed PERT method and the novel training framework. Next, Section IV provides comprehensive experimental results and analysis on RSITR. Finally, Section V concludes insights from the article and outlines potential directions for further research.

II. Related Work

This section introduces the development of RSITR and provides a review of PEFT.

A. RS Image–Text Retrieval

In the majority of prior researches on RSITR, the full fine-tuning paradigm of visual and textual models (i.e., ResNet [13], GRU [14], and Transformer [30]) has predominated. Fusion-based correlation learning model [31] captures complementary intermodality complementary information and fused features to supervise the modality-specific sub-model. An RSITR framework with semantic alignment module [12] investigates the hidden relationships between image and text modalities with attention and gating mechanisms. The asymmetric multimodal feature matching network (AMFMN) [9] employs multiscale visual self-attention to extract visually salient features and unidirectional guidance from vision to text. Lightweight multiscale cross-modal retrieval model (LW-MCR) [10] designs a distillation loss to inherit hidden representation from the teacher network [9] and boosts the performance with semi-supervised training on unlabeled data. The RSITR framework based on global and local information (GaLR) [11] demonstrates that the effective integration and mutual correction of local and global features can significantly enhance retrieval performance. Prior instruction representation

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(PIR) framework [32] incorporates prior knowledge of RS scenarios into a multimodal progressive attention encoder to optimize visual and textual representations.

Benefiting from the superior performance of the vision language model [19], the benchmark of RSITR has recently been substantially elevated to another level. Multilanguage Transformer [33] achieves remarkable retrieval performance by loading CLIP pre-trained weights and full fine-tuning on an expanded multilingual corpus. PE-RSITR [6] realizes parameter-efficient transfer learning of CLIP on RSITR through MRS-Adapter and intra-modal similarity constraints.

B. Parameter-Efficient Fine-Tuning

PEFT aims to adapt the large-scale pre-trained model W to downstream tasks by optimizing a small number of parameters ΔW, where $|\Delta W| \ll |W|$. Following the categorization criterion of [23], PEFT is divided into three families: addition tuning, specification tuning, and reparameterization tuning.

1) Addition Tuning: Addition tuning methods achieve effective transfer of large models by introducing additional modules or parameters, such as Adapter and Prompt tuning. The adapter is a small plug-and-play module inserted in the Transformer block and tuned for task adaptation. The canonical architecture of an adapter comprises a sequential arrangement of a down-projection, a nonlinear activation function, and an up-projection [34]. AdaptFormer [35] integrates the adapter in parallel with the original LayerNormalization and multilayer perceptron (MLP) of ViT blocks, facilitating an improvement on various vision tasks. Kron$\text{A}^{B}_{\text{res}}$ [27] feeds the input vector into the Kronecker product of two projection matrices and a learnable residual branch in parallel with MLP. AdaKron [36] leverages the Kronecker product between the output vectors of two down projections and is applied in sequence after MLP. Cross-modal adapter [37] adds two adapters sequentially after multihead self-attention (MHSA) and MLP within each Transformer block of CLIP's dual-branch structure, and enables cross-modal interaction by leveraging weight-sharing mechanisms. MRS-Adapter [6] firstly explores the integration of adapters into RSITR by incorporating cross-modal adapter-like modules in parallel with the LayerNormalization and MLP, further reducing the number of adapters required for cross-modal interaction. Prompt tuning introduces trainable token embeddings directly into the input features for PEFT. Visual prompt tuning (VPT) [38] prepends several visual tokens in the middle of the class token and image patch tokens to the input feature of Transformer blocks. Recognizing the sensitivity of CLIP-like vision-language models to text prompts, context optimization (CoOp) [39] converts hand-crafted text prompts into continuous learnable vectors. Despite effective fine-tuning capabilities, Adapters modify the model architecture and incur additional computational costs. Conversely, prompt tuning does not alter any module inside models but encounters difficulties in optimization.

2) Specification Tuning: Specification tuning methods fine-tune a subset of original parameters and preserve the rest frozen with the goal of sustaining performance with minimal compromise. Lee et al. [40] demonstrate that fine-tuning just one-quarter of the final layers can achieve 90% of the performance of full fine-tuning. BitFit [41] discovers that selectively updating bias terms of specific layers inside large models can achieve on-par performance with full fine-tuning of entire models. Experiments in BitFit show that even modest fine-tuning of a random subset of model parameters still yields acceptable performance. However, the criteria of specific parameter selection and the roles of various parameters in downstream tasks remain incompletely explored.

3) Reparameterization Tuning: Reparameterization tuning methods convert weight optimization into a parameter-efficient form. LoRA [24] is the most widely applied representative method for PEFT. LoRA decomposes ΔW into the matrix multiplication of two low-rank matrices A and B, where A is used for dimensionality reduction and B for restoring the dimensions. LoRA-style PEFT has attracted considerable research interest and has derived a series of studies. LoHA [42] utilizes the Hadamard product of two LoRA modules. KronA [27] simply replaces the matrix multiplication with the Kronecker product. LoKr [42] first decomposes ΔW into the Kronecker product of two matrices, and then further decomposes the first matrix like LoRA. LoRA+ [43] design an effective training approach for LoRA, assigning a smaller learning rate to matrix A and a larger learning rate to matrix B. LoRA-FA [25] freezes matrix A and solely trains matrix B, resulting in performance equivalent to LoRA but with only half the parameters. VeRA [26] modifies matrices A and B into fixed, randomly initialized matrices and simultaneously introduces a pair of adjustable vectors, thus substantially diminishing the quantity of trainable parameters. Reparameterization tuning methods provide straightforward implementations without any modification to the model architecture. While the efficacy may remain inferior to that of full fine-tuning on some complex downstream tasks.

III. Methodology

The section comprehensively introduces our proposed PERT-RaMa framework for the RSITR task. The overall architecture is illustrated in Fig. 1.

A. Preliminary: Review CLIP

Thanks to the pre-training on millions of web-crawled natural image–text pairs, CLIP has demonstrated remarkable robustness and generalization capabilities on many visual and multimodal tasks in the field of RS [6], [21], [44], [45], [46]. Thus, we also reuse the visual and textual encoders of CLIP as multimodal feature extractors. The specific pipeline of extracting image and text embeddings from CLIP is introduced as follows.

1) Visual Encoder: Given an RS image $V \in \mathbf{R}^{H\times W\times 3}$, the visual encoder of CLIP initially applies a convolution layer to perform patch partition and token embedding, generating fixed-size non-overlapping patch tokens. Following the original design in ViT [47], CLIP also inserts a classification (CLS) token at the beginning of patch tokens and adds the positional embedding to construct the RS image embedding. Then, the image embedding is fed into $l = 12$ layers of

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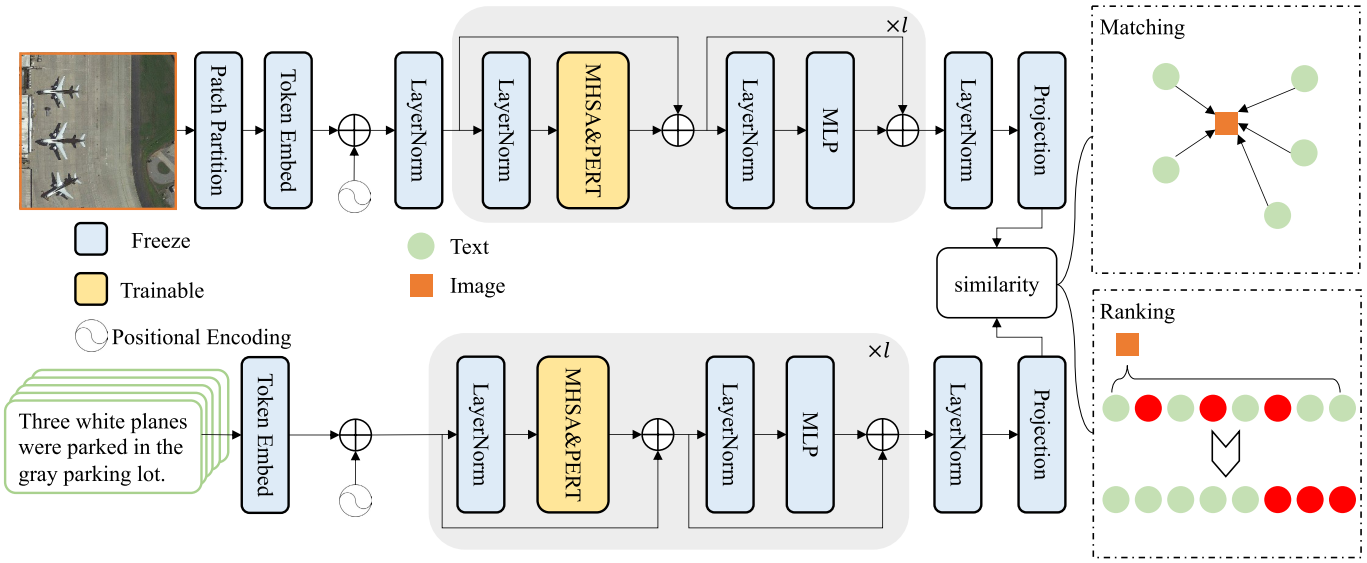


Fig. 1. Overall architecture of the proposed PERT-RaMa framework. PERT-RaMa utilizes modified CLIP as multimodal feature extractors. The modified CLIP integrates the PERT module KPLoRA into each Transformer block of visual and textual encoders. The feature similarity is optimized on the novel RaMa training method.

stacked pre-layernorm ViT blocks, which are composed of MHSA with LayerNormalization and skip-connections, and MLP also with same operations. After the linear projection into D-dimensional common embedding space, the CLS token is regarded as the global image feature.

2) Textual Encoder: Given a text caption T with arbitrary length, the textual encoder of CLIP first transforms T into unified length by essential padding or truncation, then brackets it with start-of-sentence (SOS) and end-of-sentence (EOS) tokens, and tokenizes the caption using a lower-cased byte pair encoding with a 49 152 vocab size [48]. After embedding and adding the positional embedding, the text embedding is also input into $l = 12$ layers of stacked pre-layernorm Transformer blocks. Due to the better performance of special tokens empirically, CLIP takes the EOS token of the output of the last Transformer block as the global text feature with subsequent LayerNormalization and linear projection.

3) MHSA and MLP: Considering that MHSA and MLP are closely related to the implementation of the proposed method and several comparison methods, the structures of the two modules are introduced in detail here.

MHSA first linearly transforms input feature x into query q, key k, and value v. Then, MHSA calculates the similarity between query q and key k, followed by a scaling operation to prevent gradient vanishing or exploding. Finally, a weighted sum of value v with the similarity score is computed as the output of MHSA. Mathematically, the calculation process of MHSA can be represented as

$$q, k, v = xW_q + \text{bias}_q, \quad xW_k + \text{bias}_k, \quad xW_v + \text{bias}_v \tag{1}$$

$$\text{Score} = \text{Softmax}\left(\frac{qk^{\mathrm{T}}}{\sqrt{d_k}}\right)v \tag{2}$$

$$x = \text{Score}W_o + \text{bias}_o \tag{3}$$

where d_k is the dimensionality of key k. Notably, we omit the multihead mechanism in the formulas for the sake of simplicity.

MLP is composed of two linear layers separated by Gaussian error linear units (GELUs) activation function [49]. It serves to project the input feature x into a higher-dimensional feature space before restoring it to its original dimensionality. MLP can be formulated as follows:

$$x = \text{GELU}(xW_{l1} + \text{bias}_{l1}) \tag{4}$$

$$x = xW_{l2} + \text{bias}_{l2}. \tag{5}$$

B. Kronecker Product for LoRA

In this section, we provide a detailed introduction to LoRA and the proposed KPLoRA. We then analyze and compare the intrinsic rank dimensions and parameter quantities of LoRA-style modules and our KPLoRA from the perspective of matrix rank.

LoRA [24] hypothesizes that pre-trained models are sparse and exhibit lower intrinsic rank when fine-tuned on specific downstream tasks. The formal introduction of LoRA in [24] is based on left multiplication, while implementations typically use right multiplication. To avoid any confusion, the notation of this article is aligned with the implementation. Thus, LoRA transforms the change $\Delta W \in \mathbf{R}^{D_{\text{in}} \times D_{\text{out}}}$ of weight matrix $W \in \mathbf{R}^{D_{\text{in}} \times D_{\text{out}}}$ in MHSA module into the matrix multiplication of two low-rank matrices $A \in \mathbf{R}^{D_{\text{in}} \times r}$ and $B \in \mathbf{R}^{r \times D_{\text{out}}}$, where $r \ll \min(D_{\text{in}}, D_{\text{out}})$. LoRA can be formulated as follows:

$$W \leftarrow W + \Delta W = W + \underline{AB} \tag{6}$$

where W remains frozen during the fine-tuning stage and only the underlined parameters AB are updated. Specifically, A defaults to initializing with Kaiming uniform distribution [50], while B uses zero initialization, ensuring that $\Delta W = 0$ at the first forward calculation of training. According to (6), LoRA does not introduce any additional computational cost by weight reparameterization, i.e., absorbing ΔW into W during the inference stage.

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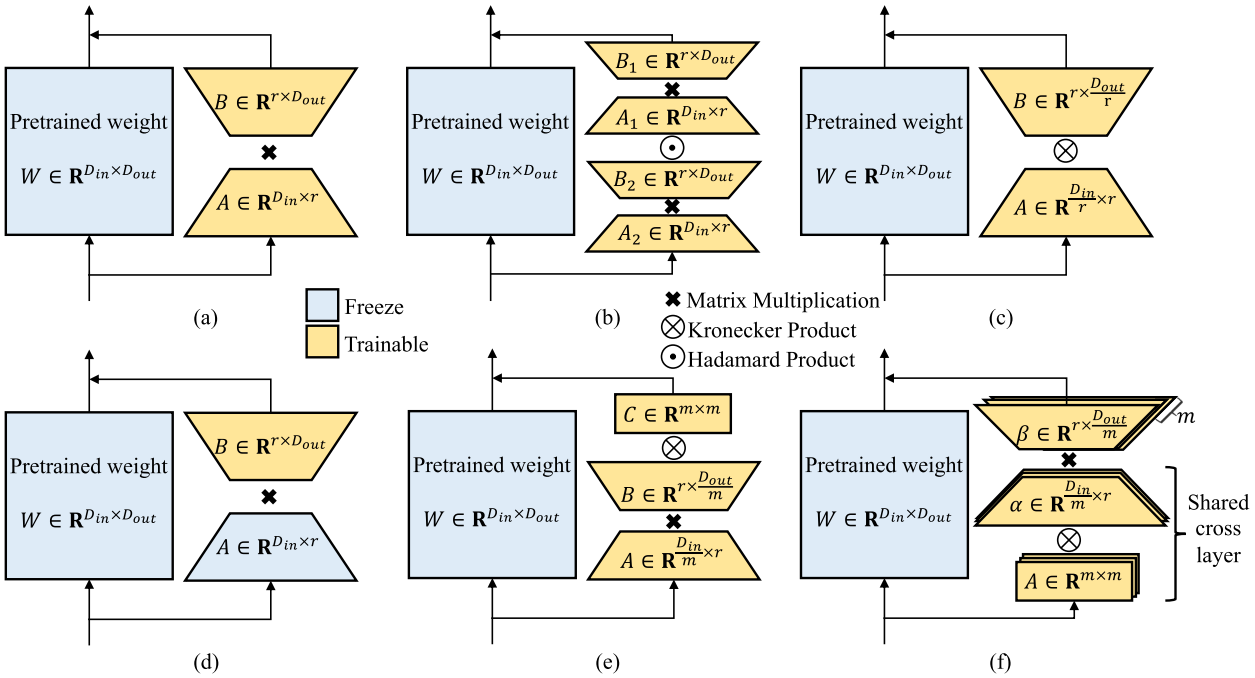


Fig. 2. Illustration of PERT modules. (a) LoRA. (b) LoHA. (c) KronA. (d) LoRA-FA. (e) LoKr. (f) KPLoRA.

The LoRA-style PERT modules [25], [27] adhere to the principle of achieving comparable performance to full fine-tuning with fewer parameters. LoHA integrates two LoRA modules A_1B_1 and A_2B_2 via the Hadamard product $\odot$ as follows:

$$W \leftarrow W + \Delta W = W + \underline{A_1B_1 \odot A_2B_2}. \tag{7}$$

LoRA-FA [25] simply freezes matrix A and only updates matrix B, enabling on-par performance with LoRA with half the parameters. LoRA-FA can be formulated as follows:

$$W \leftarrow W + \Delta W = W + A\underline{B}. \tag{8}$$

LoKr constructs a combination of the matrix multiplication and Kronecker product, which can be represented as

$$W \leftarrow W + \Delta W = W + \underline{(A \otimes B)C} \tag{9}$$

where $A \in \mathbf{R}^{(D_{\text{in}}/m) \times r}$, $B \in \mathbf{R}^{r \times (D_{\text{out}}/m)}$, and $C \in \mathbf{R}^{m \times m}$. KronA [27] can be considered a special case of LoKr when $m = 1$ and $C = \mathbf{1}$. The Kronecker product of two matrices $F \in \mathbf{R}^{a \times b}$ and $G \in \mathbf{R}^{c \times d}$ is represented by $F \otimes G$ and mathematically expressed as follows:

$$F \otimes G = \begin{bmatrix} f_{11}G & \cdots & f_{1i}G & \cdots & f_{1b}G \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ f_{a1}G & \cdots & f_{ai}G & \cdots & f_{ab}G \end{bmatrix} \tag{10}$$

where $F \otimes G \in \mathbf{R}^{ac \times bd}$ and f_{ij} is the element at the ith row and jth column of matrix F.

Motivated by Edalati et al. [27] and Karimi Mahabadi et al. [51], we propose a lightweight, plug-and-play, LoRA-style PERT module with Kronecker product for low-rank adaptation, named KPLoRA. The structures of KPLoRA and other LoRA-style modules are shown in Fig. 2. Specifically, we reform ΔW into the sum of multigroup Kronecker products, which can be mathematically defined as follows:

$$\Delta W = \sum_{i=1}^{m} A_i \otimes B_i \tag{11}$$

where $A_i \in \mathbf{R}^{m \times m}$ and $B_i \in \mathbf{R}^{(D_{\text{in}}/m) \times (D_{\text{out}}/m)}$. m denotes the number of groups. Furthermore, we decompose matrix B_i into the matrix multiplication of two low-rank matrices α_i and β_i

$$\Delta W = \sum_{i=1}^{m} A_i \otimes (\alpha_i \beta_i) \tag{12}$$

where $\alpha_i \in \mathbf{R}^{(D_{\text{in}}/m) \times r}$ and $\beta_i \in \mathbf{R}^{r \times (D_{\text{out}}/m)}$. r represents the dimension of down projection.

Following the consensus of discussed LoRA-style modules [24], [25], [27], the PERT modules are best inserted in parallel with W_q and W_v of MHSA modules, as shown in Fig. 3. Therefore, this article implements KPLoRA and reproduces other LoRA-style modules by inserting these modules in the aforementioned positions.

Moreover, a cross-layer sharing mechanism has demonstrated PEFT on RSITR task [6]. Thus, we also explore the cross-layer sharing mechanism for KPLoRA. We selectively share A_i and α_i across the Transformer layers of CLIP. A_i and α_i serve as shared parameters to learn general knowledge essential for the RSITR task, while B_i is used as layer-specific parameters to capture layer-relevant information.

1) Intrinsic Rank Analysis: The rank of a matrix is the maximal number of linearly independent vectors, denoted by $\mathcal{R}$. The rank cannot be greater than the number of columns or rows. And the rank of the matrix multiplication AB cannot exceed the smaller rank of A and B. The property is expressed as follows:

$$\mathcal{R}(AB) \leq \min(\mathcal{R}(A), \mathcal{R}(B)). \tag{13}$$

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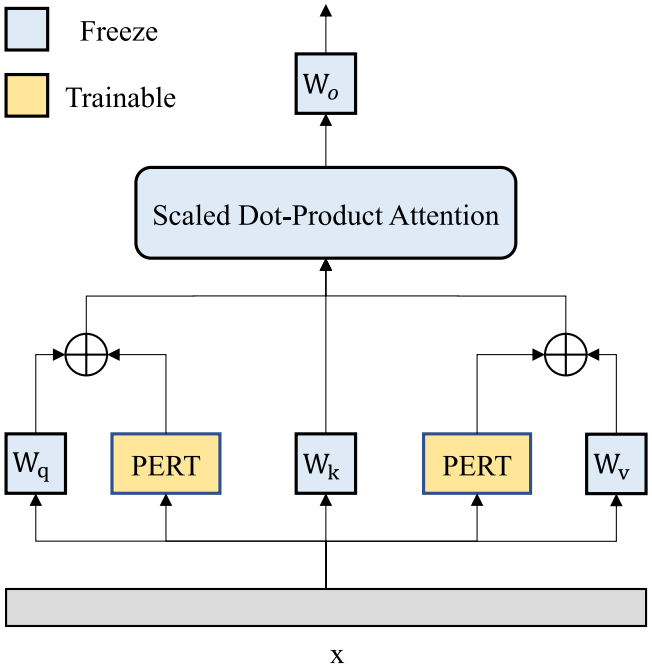


Fig. 3. Overview of PERT modules in MHSA of transformer block. Also, the multihead mechanism is ignored for clarity.

TABLE I
INTRINSIC RANK OF PERT MODULES

Method	Hyperparameter	Intrinsic Rank γ
LoRA [24]	r	r
LoHA [42]	r	r^2
KronA [27]	r	r^2
LoRA-FA [25]	r	r
LoKr [42]	m, r	mr
KPLoRA	m, r	mr

Therefore, the matrix multiplication only maintains or reduces the original rank.

The rank of the Kronecker product $A \otimes B$ is given by the product of the ranks of two matrices A and B, which can be formulated as follows:

$$\mathcal{R}(A \otimes B) = \mathcal{R}(A) \cdot \mathcal{R}(B). \tag{14}$$

The Kronecker product maintains or increases the rank in contrast. The rank of Hadamard product $A \odot B$ has a similar property and satisfies the following inequality:

$$\mathcal{R}(A \odot B) \leq \mathcal{R}(A) \cdot \mathcal{R}(B). \tag{15}$$

Table I demonstrates the relation between hyperparameters and the maximum intrinsic rank of our KPLoRA and LoRA-style modules based on (13)–(15). All hyperparameters in Table I are positive integers. In particular, LoRA essentially learns a low-rank feature subspace with the intrinsic rank that does not exceed r when training on downstream tasks. The intrinsic rank of LoRA-FA is also not greater than r. Slightly different from LoRA, the subspace of LoRA-FA is constrained by the column space of the frozen matrix A [25]. The intrinsic rank of the subspace constructed by LoHA and KronA is always a perfect square r^2, which limits its flexibility. LoKr and KPLoRA both leverage the Kronecker product and matrix multiplication, and their maximum intrinsic rank of learnable subspace is mr. Therefore, KPLoRA constructs a higher-rank feature subspace in a more flexible manner, enhancing the adaptability to downstream tasks, e.g., RSITR. Additionally, the combination of multigroup low-rank matrices effectively mitigates the overfitting problem caused by LoRA-style methods with high rank.

TABLE II
PARAMETER ANALYSIS OF REPARAMETERIZATION TUNING MODULES

Method	Hyperparam	Extra Params	
		Training	Inference
LoRA [24]	$r = \gamma$	$2\gamma lD$	0
LoHA [42]	$r = \gamma$	$4\gamma lD$	0
KronA [27]	$r = \sqrt{\gamma}$	$2lD$	0
LoRA-FA [25]	$r = \gamma$	γlD	0
LoKr [42]	$m, r = \frac{\gamma}{m}$	$m^2 l + \frac{2\gamma}{m^2} lD$	0
KPLoRA	$m, r = \frac{\gamma}{m}$	$m^3 + \frac{\gamma}{m}(l+1)D$	0

2) Parameter Analysis: We conduct an analysis of parameter count within several representative reparameterization tuning methods in Table II. Here, we take the insertion of the PERT modules with same intrinsic rank γ in $W_q \in \mathbf{R}^{D_{\text{in}} \times D_{\text{out}}}$ of l-layer Transformer blocks as an example to calculate the parameters. As for CLIP, D_{in} always equals D_{out} in both visual and textual encoders, so D is used to represent the dimension of W_q in Table II.

In the setting of same intrinsic rank γ, LoRA has the trainable parameters $2\gamma lD$, while LoHA doubles the parameter count of LoRA and LoRA-FA reduces that by half. The parameters of KronA only depend on the size of the original weight matrix and the number of inserted KronA modules. KPLoRA approximately reduces the parameters to $(2/m)$ of that of LoRA. When γ is fixed, the parameter quantity of the first three PERT modules remains constant. However, the number of parameters of LoKr and KPLoRA may vary due to different combinations of m and r. The specific values of hyperparameters refer to Section IV-D.

These discussed LoRA-style methods and our proposed KPLoRA do not involve any nonlinear transformations or input-dependent computations. These PERT modules can be merged into original pre-trained weights through element-wise addition, without introducing any extra parameters during the inference stage.

C. Ranking and Matching

In this section, we first analyze the inconsistency between the task modeling of existing RSITR methods and evaluation benchmarks, as well as the substantial redundant computations in the common training method. This article names the common training method BiMa. Then, we introduce an asymmetric task modeling that is more aligned with the characteristics of RSITR data. Furthermore, we propose a novel training method named RaMa, as shown in Fig. 4.

RSITR consists of two subtasks: image–text retrieval (*i2t*) and text–image retrieval (*t2i*). The common task definition of RSITR is shown as follows.

Task definition: Given a set of RS images $\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$ and text captions $\mathcal{T} = \{t_1, t_2, \ldots, t_m\}$, the

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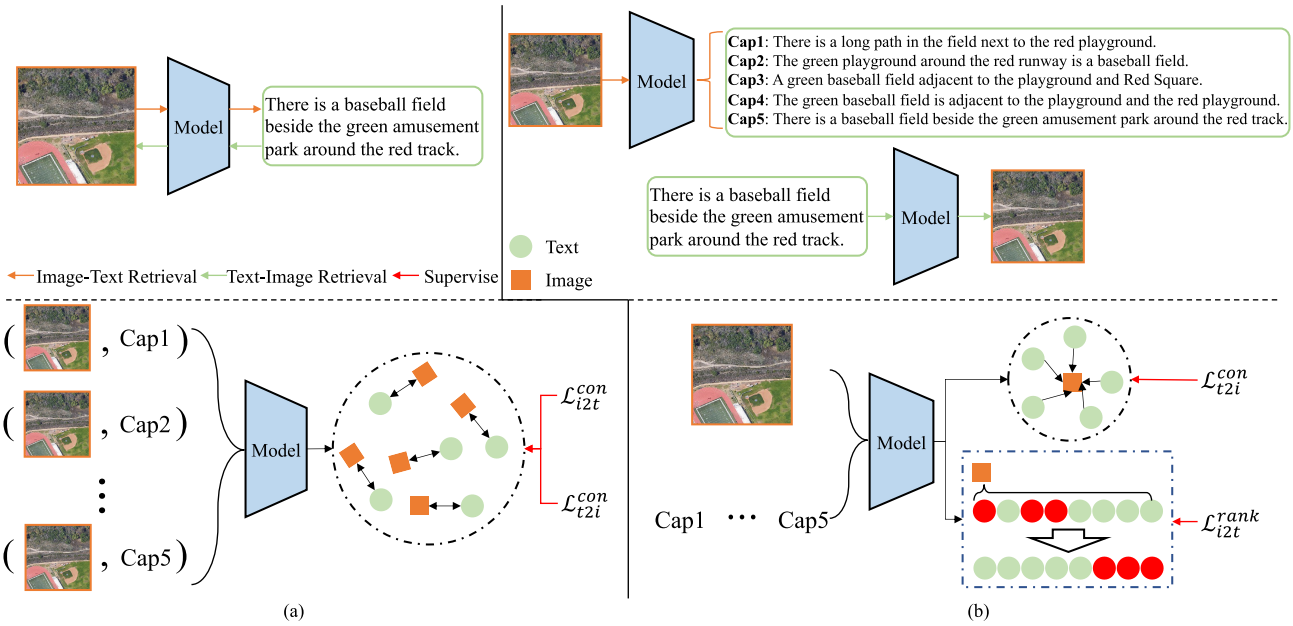


Fig. 4. Intuitive comparison of two training methods. (a) BiMa and (b) RaMa. The portion above the dashed line represents task modeling and understanding, while the part below the dashed line shows the training pipeline.

RSITR task aims to return the similar image with a text query, and vice versa.

Based on the above task definition, existing RSITR methods [6], [9], [11] usually assumes that *i2t* and *t2i* are symmetric and model them as a BiMa task, as shown in Fig. 4. The BiMa task aims to retrieve the best matched sample in one modality to the given query in the other modality. In practice, RS images are acquired by the Earth observation of satellites, while text captions are provided based on human requirements. Human understanding of RS images is diverse, and human language is varied and arbitrary. Therefore, current evaluation benchmarks [9], [28], [29] typically annotate five text captions for each RS image, aiming to be closer to the real-world application. The *i2t* subtask actually aims to recall five text captions to one RS image query, while the *t2i* subtask still matches only one ground-truth RS image to one text query. The inconsistency between task modeling and evaluation benchmarks leads to a gap between learning objectives and retrieval requests.

Moreover, existing RSITR methods optimize the retrieval model with the BiMa training, as shown in Fig. 4, which is named BiMa in this article. BiMa individually pairs each RS image with five text captions in an epoch, and input these pairs into the model for contrastive learning. BiMa repeatedly extracts features of each RS image several times, which results in unnecessary redundant computations and slow training speed.

To mitigate the inconsistencies, we modify the task modeling of RSITR. As shown in Fig. 4, we consider the *i2t* subtask as a ranking problem that is dedicated to maintaining a better sequence of retrieved results for the RS image query. The *t2i* subtask remains a matching problem and still aims to find the best-matched RS image for the text query. Corresponding to the asymmetric task modeling, we redesign a novel training method, named RaMa.

The pipeline of RaMa is shown in Algorithm 1. During the training stage, RaMa combines each RS image and five ground-truth text captions as an instance of a mini-batch. RaMa enables the visual encoder to extract features only once for each RS image, significantly removing the redundant calculations per epoch. RaMa implements an asymmetric joint loss function to guide the retrieval model to converge to the optimal state. The loss function is described in detail in Section III-D.

Algorithm 1 Pipeline of RaMa

Input: A mini-batch $\mathcal{B}$ consists of $(v_i, t_i^1, t_i^2, t_i^3, t_i^4, t_i^5), i = 1, \ldots, \mathcal{B}$

Extract features $F_{\mathcal{V}} = \{f_{v_i}\}$ of RS image v_i with visual encoder

Extract features $F_{\mathcal{T}} = \left\{f_{t_i^1}, \ldots, f_{t_i^5}\right\}$ of texts $(t_i^1, \ldots, t_i^5)$ with textual encoder

Calculate cosine similarity $\mathcal{S}_{i2t} = cos(F_{\mathcal{V}}, F_{\mathcal{T}})$ and $\mathcal{S}_{t2i} = cos(F_{\mathcal{T}}, F_{\mathcal{V}})$

Optimize $\mathcal{S}_{i2t}$ with rank loss $\mathcal{L}^{rank}$

Optimize $\mathcal{S}_{t2i}$ with contrastive loss $\mathcal{L}^{con}$

Output: $\mathcal{S}_{i2t}$ and $\mathcal{S}_{t2i}$

D. Loss Function

Existing RSITR methods [6], [9], [11] typically utilize symmetric triplet loss [52] corresponding to the BiMa training method. The symmetric triplet loss brings matched image–text pairs closer while pushing unmatched image–text pairs further apart

$$\begin{aligned}\mathcal{L}^{\text{BiMa}}(v,t) &= \mathcal{L}_{i2t}^{\text{BiMa}}(v,t) + \mathcal{L}_{t2i}^{\text{BiMa}}(t,v) \\ &= \sum_{\hat{t}} \max[0, \mu - s_{v,t} + s_{v,\hat{t}}] \\ &\quad + \sum_{\hat{v}} \max[0, \mu - s_{t,v} + s_{t,\hat{v}}] \end{aligned} \tag{16}$$

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TABLE III
STATISTICS FOR THREE EVALUATION DATASETS

Method	Year	Image Quantity	Captions/image	Resolution
RSICD	2017	10921	5	224×224
RSITMD	2022	4743	5	256×256
UCM	2016	2100	5	256×256

where μ is the margin of matched pairs and unmatched pairs. s generally denotes the cosine similarity. $\hat{\cdot}$ represents the unmatched sample. Instead of the symmetric triplet loss, we employ an appropriate asymmetric loss function to guide the learning of asymmetric subtasks corresponding to the novel training method RaMa.

1) Text–Image Retrieval: RaMa still treats the *t2i* subtask as a matching problem. We preserve the triplet loss to supervise the retrieval model in searching the uniquely matched RS image for each text query

$$\mathcal{L}_{t2i}^{\mathrm{RaMa}}(t) = \sum_{\hat{v}} \max[0, \mu - s_{t,v} + s_{t,\hat{v}}] \tag{17}$$

where μ is set to 0.2 in the article referring to [6], [10], and [11].

2) Image–Text Retrieval: RaMa transforms the *i2t* subtask from a matching problem into a ranking problem. Inspired by Recall@k Surrogate Loss [53] designed for image retrieval task, we modify the loss function to accommodate the *i2t* subtask. Recall@k Surrogate Loss directly optimizes the differentiable version of Recall at K and maintains a better retrieval sequence, which can be formulated as follows:

$$\mathcal{L}^{k}(v) = 1 - \frac{\sum_{t \in P_v} \sigma_{\tau_1}\left(k - 1 - \sum_{z \in \Omega_v, z \neq t} \sigma_{\tau_1}\left(s_{v,z} - s_{v,t}\right)\right)}{|P_v|} \tag{18}$$

$$\mathcal{L}_{i2t}^{\mathrm{RaMa}}(v) = \frac{1}{|K|} \sum_{k \in K} \mathcal{L}^{k}(v) \tag{19}$$

where P_v is the set of matched text captions to the image query v. Ω_v denotes the set of all candidate text captions to v. σ represents the sigmoid function and the temperature τ. controls the smoothness. k means the Top-k retrieved results and K is the set of different values of k. Following the default settings in [53], $\tau_1 = 1$, $\tau = 0.01$, and $K = \{1, 2, 4, 8\}$. For more details about Recall@k Surrogate Loss, refer to [53].

The overall loss function is a linear combination of the loss functions of *t2i* and *i2t* subtasks

$$\mathcal{L}^{\mathrm{RaMa}}(v, t) = \mathcal{L}_{i2t}^{\mathrm{RaMa}}(v) + \mathcal{L}_{t2i}^{\mathrm{RaMa}}(t). \tag{20}$$

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: The article evaluates our proposed method on three widely used RSITR datasets, including RS image caption dataset (RSICD) [28], RS image–text matching dataset (RSITMD) [9], and UC Merced land use (UCM) dataset [29]. The detailed information of three datasets is shown in Table III.

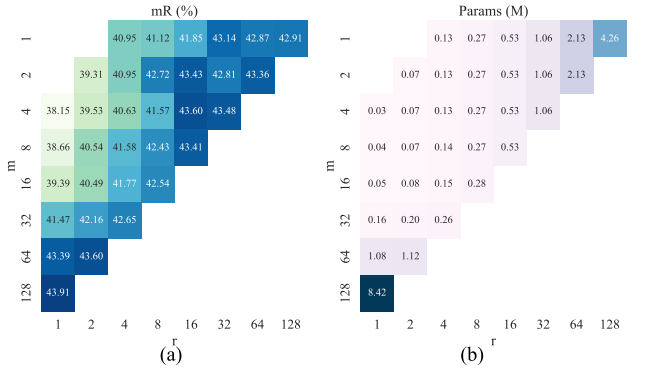


Fig. 5. Ablations of hyperparameters m and r. (a) and (b) mR and trainable parameters of PERT-RaMa with different values of m and r.

2) Evaluation Metrics: To evaluate and fairly compare the effectiveness of our proposed method for RSITR, we employ two metrics: Recall at K ($R@K$) and mean Recall (mR), following [6], [9], and [11]. In the context of RSITR, $R@K$ measures the proportion of relevant text captions retrieved among the top-K results that accurately corresponds to the RS image query or vice versa. By varying the value of K ($K = 1$, 5, and 10), we assess the retrieval performance at different levels of result granularity for *i2t* and *t2i* subtasks. mR provides an aggregate measure of the average recall and offers insights into the overall retrieval performance. The two metrics are formulated as follows:

$$R@K = \sum_{Q} \frac{\mathrm{TP}@K}{\mathrm{TP}@K + \mathrm{FN}@K} \tag{21}$$

$$\mathrm{mR} = \frac{1}{2}\left(\underbrace{\frac{1}{3}\sum_{K=1,5,10} R@K}_{i t 2\,\mathrm{mR}} + \underbrace{\frac{1}{3}\sum_{K=1,5,10} R@K}_{t 2 i\,m R}\right). \tag{22}$$

3) Implementation Details: To maintain consistency with the experimental setup of previous works [6], [9], [11], we also employ k-fold cross-validation to accurately estimate model performance, where $k = 5$. The average results of k-fold cross-validation are reported in this article. Concretely, 80%, 10%, and 10% of the dataset are used for training, validation, and testing. The intrinsic rank of KPLoRA module varies in {4, 8, 16, 32, 64, 128}. The specific values of hyperparameters m and r refer to Section IV-D. According to the default image resolution of CLIP [19], all RS images are resized into 224×224. During the training phase, the AdamW optimizer is utilized to iteratively adjust the trainable parameters with a batch size of 16 for 30 epochs. Model training begins with a learning rate of 0.0002, which decays by a factor of 0.7 every 20 epoch. All experiments are conducted on one NVIDIA RTX3090 GPU (24 GB).

B. Comparisons With SOTA Methods

In this section, we compare our proposed method with representative RSITR methods of full fine-tuning paradigm and PEFT paradigm.

1) Full Fine-Tuning Paradigm: Full fine-tuning paradigm optimizes all parameters of the pre-trained model. Following prior researches [6], [9], VSE++ [54], LW-MCR [10],

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TABLE IV
QUANTITATIVE RESULTS OF SOTA METHODS ON RSICD TEST SET

Method	Backbone Visual Encoder, Text Encoder	Trainable Params	Image-Text Retrieval			Text-Image Retrieval			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
Full Fine-tuning Paradigm									
VSE++ [54]	ResNet18, GRU	15.78M	3.38	9.51	17.46	2.82	11.32	18.10	10.43
LW-MCR [10]	SqueezeNet, Group Convolution	1.65M	4.39	13.35	20.29	4.30	18.85	32.34	15.59
AMFMN [9]	ResNet18, biGRU	35.94M	5.39	15.08	23.40	4.90	18.28	31.44	16.42
GaLR [11]	ResNet18, biGRU	46.89M	6.59	19.85	31.04	4.69	19.48	32.13	18.96
PIR [32]	Swin Transformer+ResNet-50, Bert	161M	9.88	27.26	39.16	6.97	24.56	38.92	24.46
Multilanguage Transformer [33]	CLIP(ViT-B-32)	151M	10.70	29.64	41.53	9.14	28.96	44.59	27.42
Zero-shot CLIP [19]	CLIP(ViT-B-32)	0.00M	6.77	15.37	23.15	5.01	15.75	24.21	15.04
Full fine-tuned CLIP [1]	CLIP(ViT-B-32)	151M	13.54	30.83	43.46	11.55	33.14	49.83	30.39
Parameter-efficient Fine-tuning Paradigm									
CoOp [39]	CLIP(ViT-B-32)	0.10M	6.32	15.89	27.60	4.31	17.89	31.73	17.29
VPT [38]	CLIP(ViT-B-32)	0.46M	7.23	19.40	30.77	6.94	25.26	40.73	21.72
Cross-Modal Adapter [37]	CLIP(ViT-B-32)	0.16M	11.18	27.31	40.62	9.57	30.74	48.36	27.96
AdaptFormer [35]	CLIP(ViT-B-32)	0.17M	12.46	28.49	41.86	9.09	29.89	46.81	28.10
$\text{KronA}^{\text{B}}_{\text{res}}$ [27]	CLIP(ViT-B-32)	0.03M	7.65	20.81	32.75	6.08	22.58	38.13	21.33
AdaKron [36]	CLIP(ViT-B-32)	1.00M	10.12	26.99	39.78	9.02	30.12	46.45	27.08
PE-RSITR [6]	CLIP(ViT-B-32)	0.16M	14.13	31.51	44.78	11.63	33.92	50.73	31.12
Lee et al. [40]	CLIP(ViT-B-32)	40.96M	6.62	20.07	31.22	5.61	21.32	36.75	20.27
BitFit [41]	CLIP(ViT-B-32)	0.03M	10.32	24.74	37.13	8.28	28.11	43.97	25.42
LoRA [24]	CLIP(ViT-B-32)	0.49M	13.19	31.47	45.00	10.43	31.85	48.71	30.11
LoHA [42]	CLIP(ViT-B-32)	0.98M	11.51	28.33	42.01	10.33	32.79	50.08	29.17
KronA [27]	CLIP(ViT-B-32)	0.06M	11.02	26.97	40.91	9.60	31.30	47.99	27.96
LoRA-FA [25]	CLIP(ViT-B-32)	0.25M	11.80	30.21	42.96	9.91	31.27	47.85	29.00
LoKr [42]	CLIP(ViT-B-32)	0.49M	11.86	29.33	43.73	10.48	32.96	49.93	29.72
PERT-RaMa (Ours)	CLIP(ViT-B-32)	0.53M	12.97	31.53	45.20	11.47	34.04	51.46	31.11

AMFMN [9], GaLR [11], and Multilanguage Transformer [33] are compared across three datasets. Moreover, we also add the latest RSITR method, PIR [32]. We report the results of zero-shot and full fine-tuned CLIP [6] in this section.

2) PEFT Paradigm: PEFT paradigm adjusts a small subset of the pre-trained model's parameters, including additional tuning, specification tuning, and reparameterization tuning.

1) *Addition tuning:* We compare several representative approaches, including prompt tuning (CoOp [39], VPT [38]), and adapter methods (cross-modal adapter [37], AdaptFormer [35], PE-RSITR [6]). PE-RSITR is the state-of-the-art (SOTA) method in PEFT paradigm. Practically, we directly report the quantitative results of addition tuning methods reproduced by Yuan et al. [6]. We further incorporate $\text{Kron}^{B}_{\text{res}}$ [27] and AdaKron [36], both based on the Kronecker product, as comparison methods.
2) *Specification tuning:* We reproduce the idea of Lee et al. [40] through only fine-tuning the last four transformer blocks and linear projection layer in CLIP. BitFit [41] only optimizes bias terms of particular layers in the original model, e.g., Linear and LayerNormalization layers. In comparative experiments in the article, we also implement BitFit [41] in CLIP [19] for RSITR task with trainable bias_q and bias_{l2} by default.
3) *Reparameterization tuning:* It is worth noticed that LoRA [24], LoHA [42], KronA [27], LoRA-FA [25], and LoKr [42] are not specifically designed for CLIP or RSITR. Therefore, we reproduce these methods on the proposed PERT-RaMa framework with the aforementioned implementation details on three RSITR datasets. These compared LoRA-style methods are inserted only into W_q and W_v of every transformer block in visual and textual encoding branches. The intrinsic rank γ also varies in $\{4, 8, 16, 32, 64, 128\}$ and the non-final low-rank matrices are initialized with Xavier Normal distribution. Specifically, based on the analysis of intrinsic rank in Section III-B, the intrinsic rank γ of LoHA and KronA must be a perfect square number, so γ varies in $\{4, 16, 64\}$. Moreover, LoKr adopts the same m and r as the proposed KPLoRA under different intrinsic ranks. Tables IV–VI only report the best results of PERT methods on three datasets. We also provide the performance of these methods with different intrinsic ranks in Table VII.

C. Quantitative Results Analysis

1) Comparison on Three Datasets: Tables IV–VI present quantitative results on the test set of RSICD, RSITMD, and UCM, respectively. Our method efficiently transfers CLIP to RSITR, outperforming other full fine-tuning methods without CLIP by approximately 5%–10%. AMFMN, GaLR, and PIR are specifically designed for RSITR, which fully considering the global and local information of RS images. However, these methods have not obtained satisfactory retrieval performance due to the limitation of poor representation ability of backbones. Zero-shot CLIP has achieved comparable or superior retrieval performance to VSE++, demonstrating CLIP's generalization capabilities. The application of CLIP has made a leap for the RSITR benchmark. Multilanguage Transformer replaces the last linear projection with global average pooling and loads pre-trained weights of CLIP, resulting in inferior recall to full fine-tuned CLIP. Our method is superior to full fine-tuned CLIP on the largest dataset RSICD and smallest dataset UCM with only 0.53M trainable parameters. Our method demonstrates performance comparable to full

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TABLE V
QUANTITATIVE RESULTS OF SOTA METHODS ON RSITMD TEST SET

Method	Backbone Visual Encoder, Text Encoder	Trainable Params	Image-Text Retrieval			Text-Image Retrieval			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
Full Fine-tuning Paradigm									
VSE++ [54]	ResNet18, GRU	15.78M	10.38	27.65	39.60	7.79	24.87	38.67	24.83
LW-MCR [10]	SqueezeNet, Group Convolution	1.65M	9.73	26.77	37.61	9.25	34.07	54.03	28.58
AMFMN [9]	ResNet18, biGRU	35.94M	10.63	24.78	41.81	11.51	34.69	54.87	29.72
GaLR [11]	ResNet18, biGRU	46.89M	14.82	31.64	42.48	11.15	36.68	51.68	31.41
PIR [32]	Swin Transformer+ResNet-50, Bert	161M	18.14	41.15	52.88	12.17	41.68	63.41	38.24
Multilanguage Transformer [33]	CLIP(ViT-B-32)	151M	19.69	40.26	54.42	17.61	49.73	66.59	41.38
Zero-shot CLIP [19]	CLIP(ViT-B-32)	0.00M	9.29	26.33	37.39	7.79	23.67	38.89	23.89
Full fine-tuned CLIP [6]	CLIP(ViT-B-32)	151M	24.16	47.12	61.28	20.40	50.53	68.54	45.33
Parameter-efficient Fine-tuning Paradigm									
CoOp [39]	CLIP(ViT-B-32)	0.10M	12.19	30.69	42.82	9.16	33.85	54.35	30.51
VPT [38]	CLIP(ViT-B-32)	0.46M	14.98	32.05	40.15	15.97	41.5	60.35	34.14
Cross-Modal Adapter [37]	CLIP(ViT-B-32)	0.16M	18.16	36.08	48.72	16.31	44.33	64.75	38.06
AdaptFormer [35]	CLIP(ViT-B-32)	0.17M	16.71	30.16	42.91	14.27	41.53	61.46	34.81
KronA$^{\mathrm{B}}_{\mathrm{res}}$ [27]	CLIP(ViT-B-32)	0.03M	12.88	30.44	44.51	11.19	38.93	58.94	32.81
AdaKron [36]	CLIP(ViT-B-32)	1.00M	15.71	34.87	47.70	13.80	43.38	63.97	36.57
PE-RSITR [6]	CLIP(ViT-B-32)	0.16M	23.67	44.07	60.36	20.10	50.63	67.97	44.47
Lee et al. [40]	CLIP(ViT-B-32)	40.96M	14.25	34.25	47.70	12.40	41.01	62.01	35.27
BitFit [41]	CLIP(ViT-B-32)	0.03M	15.66	35.13	48.41	14.93	42.37	62.05	36.43
LoRA [24]	CLIP(ViT-B-32)	1.97M	21.19	42.26	56.90	17.58	50.03	68.74	42.78
LoHA [42]	CLIP(ViT-B-32)	0.98M	18.10	40.09	55.09	17.58	47.50	67.69	41.01
KronA [27]	CLIP(ViT-B-32)	0.06M	16.77	38.54	51.81	15.42	45.88	65.32	38.96
LoRA-FA [25]	CLIP(ViT-B-32)	0.98M	19.65	41.99	56.06	16.73	48.53	67.78	41.79
LoKr [42]	CLIP(ViT-B-32)	0.49M	17.65	40.31	53.76	16.88	47.38	67.22	40.53
PERT-RaMa (Ours)	CLIP(ViT-B-32)	0.53M	20.66	44.16	58.94	17.95	50.81	69.08	43.60

TABLE VI
QUANTITATIVE RESULTS OF SOTA METHODS ON UCM TEST SET

Method	Backbone Visual Encoder, Text Encoder	Trainable Params	Image-Text Retrieval			Text-Image Retrieval			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
Full Fine-tuning Paradigm									
VSE++ [54]	ResNet18, GRU	15.78M	12.38	44.76	65.71	10.10	31.80	56.85	36.93
LW-MCR [10]	SqueezeNet, Group Convolution	1.65M	18.10	47.14	63.81	13.14	50.38	79.52	45.35
AMFMN [9]	ResNet18, biGRU	35.94M	16.67	45.71	68.57	12.86	53.24	79.43	46.08
Multilanguage Transformer [33]	CLIP(ViT-B-32)	151M	19.04	53.33	77,61	19.33	64.00	91.42	54.12
Zero-shot CLIP [19]	CLIP(ViT-B-32)	0.00M	10.95	34.76	55.71	7.33	35.14	54.57	33.08
Full fine-tuned CLIP [6]	CLIP(ViT-B-32)	151M	17.14	55.24	79.52	13.90	56.95	91.81	52.43
Parameter-efficient Fine-tuning Paradigm									
CoOp [39]	CLIP(ViT-B-32)	0.10M	7.19	42.76	74.19	9.78	48.34	87.41	44.95
VPT [38]	CLIP(ViT-B-32)	0.46M	8.86	47.81	78.62	13.64	55.61	94.05	49.77
Cross-Modal Adapter [37]	CLIP(ViT-B-32)	0.16M	13.77	50.57	81.41	17.43	61.45	95.48	53.62
AdaptFormer [35]	CLIP(ViT-B-32)	0.17M	16.92	54.39	77.46	18.74	63.49	91.16	53.69
KronA$^{\mathrm{B}}_{\mathrm{res}}$ [27]	CLIP(ViT-B-32)	0.03M	15.05	51.24	76.10	15.52	59.81	92.74	51.74
AdaKron [36]	CLIP(ViT-B-32)	1.00M	14.57	49.81	73.90	13.39	55.73	89.52	49.49
PE-RSITR [6]	CLIP(ViT-B-32)	0.16M	22.71	55.81	80.33	18.82	62.84	93.72	55.71
Lee et al. [40]	CLIP(ViT-B-32)	40.96M	14.29	50.76	75.24	15.03	61.90	94.15	51.90
BitFit [41]	CLIP(ViT-B-32)	0.03M	15.62	52.57	79.43	15.62	60.88	95.83	53.32
LoRA [24]	CLIP(ViT-B-32)	0.49M	21.71	58.10	81.24	18.08	65.09	95.26	56.58
LoHA [42]	CLIP(ViT-B-32)	0.25M	18.19	57.90	87.05	16.59	62.29	96.06	56.35
KronA [27]	CLIP(ViT-B-32)	0.06M	19.81	56.67	83.05	17.14	64.50	96.25	56.23
LoRA-FA [25]	CLIP(ViT-B-32)	0.25M	18.38	58.19	81.33	18.70	65.12	94.10	55.97
LoKr [42]	CLIP(ViT-B-32)	0.49M	19.33	58.10	82.29	17.12	64.21	95.83	56.15
PERT-RaMa (Ours)	CLIP(ViT-B-32)	0.53M	19.05	59.05	84.38	18.82	64.90	94.82	56.83

fine-tuned CLIP on the RSITMD dataset. Similar to other PERT methods, our method updates only a small portion of specific low-rank matrices within the pre-trained CLIP, significantly reducing the computational cost. However, this approach may limit the model's capacity to make substantial weight adjustments, which can be essential for complex datasets with a significantly different data distribution from the original training set. In contrast, full fine-tuning enables more comprehensive weight modifications, allowing the model to better adapt to complex data characteristics. Nevertheless, the results demonstrate the high generalization capability of our method, achieving a balance between efficiency and performance without extensive weight modifications.

We further compare PEFT methods. The poor performance of representative prompt tuning methods, CoOp and VPT, across all datasets indicates that optimizing a single modality alone cannot effectively eliminate domain differences. And optimizing the visual encoder is more critical than the textual encoder for transferring CLIP to RSITR, aligning with the analysis from the final ablation experiment. Compared with the leading adapter method PE-RSITR, our method achieves superior performance on the UCM dataset and comparable recall

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TABLE VII

ANALYSIS OF INTRINSIC RANK ON RSITMD, RSITMD, AND UCM DATASETS. THE SUBSCRIPT NUMBERS OF KPLORA ARE THE VALUES OF m AND r

Intrinsic Rank γ	Method	Params	RSICD			RSITMD			UCM		
			i2t mR	t2i mR	mR	i2t mR	t2i mR	mR	i2t mR	t2i mR	mR
4	LoRA	0.25M	30.06	29.75	29.90	36.37	42.63	39.50	52.98	59.35	56.17
	LoHA	0.25M	26.53	29.06	27.79	34.38	41.95	38.16	54.38	58.31	56.35
	KronA	0.06M	26.30	29.63	27.96	34.16	40.31	37.23	50.98	58.35	54.67
	LoRA-FA	0.12M	28.86	28.61	28.74	35.66	42.07	38.87	51.84	59.17	55.51
	LoKr	0.25M	27.20	29.78	28.49	36.42	42.77	39.59	52.70	58.45	55.57
	KPLoRA$_{\{1,4\}}$	0.13M	28.53	31.00	29.77	38.05	43.84	40.95	53.94	59.23	56.58
8	LoRA	0.49M	29.89	30.33	30.11	37.24	44.24	40.74	53.68	59.47	56.58
	LoRA-FA	0.25M	28.75	29.13	28.94	36.37	43.21	39.79	51.71	59.21	55.46
	LoKr	0.49M	28.31	31.13	29.72	37.24	43.83	40.54	53.30	58.91	56.10
	KPLoRA$_{\{1,8\}}$	0.27M	29.30	31.50	30.40	38.16	44.07	41.12	54.13	59.31	56.72
16	LoRA	0.98M	29.86	30.08	29.97	39.16	44.89	42.02	53.43	59.60	56.51
	LoHA	0.49M	28.00	30.14	29.07	35.65	43.04	39.35	53.56	58.74	56.15
	KronA	0.06M	26.53	26.88	26.70	35.71	42.20	38.96	52.22	58.84	55.53
	LoRA-FA	0.49M	28.33	29.68	29.00	38.54	44.44	41.49	50.19	59.64	54.92
	LoKr	0.25M	27.36	29.75	28.55	35.44	42.51	38.98	51.87	58.41	55.14
	KPLoRA$_{\{2,8\}}$	0.27M	29.64	31.50	30.57	40.28	45.17	42.72	53.71	59.43	56.57
32	LoRA	1.97M	27.73	30.43	29.08	40.12	45.45	42.78	52.44	58.58	55.51
	LoRA-FA	0.98M	26.21	29.09	27.65	39.23	44.35	41.79	52.63	59.31	55.97
	LoKr	0.49M	27.75	31.03	29.39	36.58	43.27	39.92	53.24	59.05	56.15
	KPLoRA$_{\{2,16\}}$	0.53M	28.79	31.45	30.12	41.58	45.29	43.43	53.46	59.02	56.24
64	LoRA	3.93M	24.89	28.24	26.57	38.29	44.86	41.58	50.44	58.50	54.47
	LoHA	0.98M	27.28	31.06	29.17	37.76	44.26	41.01	53.17	59.21	56.19
	KronA	0.06M	27.42	26.46	26.94	34.17	39.66	36.92	53.17	59.30	56.23
	LoRA-FA	1.97M	24.06	27.07	25.56	37.20	43.46	40.33	50.79	58.95	54.87
	LoKr	0.25M	27.09	29.30	28.19	34.81	41.19	38.00	52.44	58.47	55.46
	KPLoRA$_{\{4,16\}}$	0.53M	29.47	32.09	30.78	41.25	45.95	43.60	54.16	59.51	56.83
128	LoRA	7.86M	21.83	25.07	23.45	36.24	42.60	39.42	49.17	58.85	54.01
	LoRA-FA	3.93M	21.01	23.95	22.48	35.44	40.37	37.90	49.90	58.47	54.19
	LoKr	0.13M	25.38	27.92	26.65	34.09	39.70	36.89	51.02	57.62	54.32
	KPLoRA$_{\{8,16\}}$	0.53M	29.90	32.32	31.11	40.84	45.97	43.41	53.87	59.47	56.67

on the RSICD and RSITMD datasets. PE-RSITR leverages intra-modal constraints based on a random dropout operation, which enhances the mR on the RSITMD dataset from 40.91 to 44.47 as noted in the original paper [6]. However, implementing the intra-modal constraint mechanism requires repeated forward propagations during training, leading to increased computational demands and reduced efficiency. Our method adopts the proposed PERT-RaMa framework and PERT module KPLoRA, which effectively reduces computational costs on the training stage and preserves the original architecture of CLIP during inference with a minor trade-off in recall.

Notably, our PERT-RaMa method performs slightly worse than other methods in terms of R@1 across all datasets. However, this outcome is consistent with the task modeling approach presented in this article. In the existing datasets, where each image is associated with multiple text captions, we reformulate the *i2t* task as a ranking problem rather than a strict matching task. The goal of ranking optimization is not solely to identify the best match but to construct an improved retrieval sequence, ensuring that more target samples are ranked higher in the retrieval sequence. As a result, our method outperforms other methods in R@5 and R@10 metrics, demonstrating its effectiveness in ranking-based retrieval. For *t2i* task, which is still modeled as a matching task, our method achieves comparable R@1 performance on the UCM and RSICD datasets to other methods. However, due to the mutual influence between the *i2t* and *t2i* tasks during the optimization process, a slight decrease in *t2i* R@1 is observed on the RSITMD dataset. This minor trade-off is a consequence of the joint optimization of both tasks and highlights the balance between RaMa objectives in our framework.

For specification tuning, adjusting parameters across more layers is more suitable for RSITR than adjusting more parameters within each layer. Lee et al. [40] optimize the last one-fourth transformer blocks and last linear projection, while BitFit only unfreezing bias$_q$ and bias$_{l2}$ of all transformer blocks. However, manually selecting specific parameters for training that are highly relevant to downstream tasks can be challenging, making CLIP unable to effectively adapt to the RS domain.

Moreover, reparameterization tuning methods reproduced with PERT-RaMa framework all achieve competitive performance. The proposed KPLoRA with PERT-RaMa framework obviously performs better than various LoRA-style methods considering both recall and parameters across all datasets.

2) Comparison With Different Intrinsic Ranks: Table VII provides a comprehensive comparison between KPLoRA and other LoRA-style modules with different intrinsic ranks. Overall, KPLoRA performs the best in almost evaluation metrics on three datasets among all values of the intrinsic rank. As the intrinsic rank increases from 4 to 32, the mR of LoRA on RSITMD improves from 39.50 to 42.78, while that of LoRA-FA varies from 38.87 to 41.79. However, the mR of LoRA and LoRA-FA, respectively, reduces to 39.42 and 37.90 when the intrinsic rank further increases to 128. LoRA and LoRA-FA struggle to overcome domain differences and learning the diversity of RSITR with small intrinsic rank, but clearly exhibit overfitting and recall degradation with

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large intrinsic rank. Conversely, KPLoRA can improve or maintain retrieval performance as intrinsic rank increases. KPLoRA decomposes the matrix with large rank into the sum of multigroup low-rank matrices, effectively alleviating the overfitting problem. Moreover, KPLoRA, KronA, and LoKr all are developed based on the Kronecker product, while the joint learning of multigroup matrices A, α, and β allows KPLoRA to be more flexible in the intrinsic rank and extensibility from natural scenes to RS scenes.

D. Ablation Study

In this section, we conduct comprehensive ablation studies to evaluate the effectiveness and efficiency of primary components of the PERT-RaMa framework, i.e., hyperparameters, training method, matrix initialization, and cross-layer sharing mechanism. Considering RSITMD is a more fine-grained and challenging RSITR dataset, we perform detailed ablation experiments on RSITMD dataset with the evaluation metrics of *i2t* mR, *t2i* mR, and mR.

1) Analysis on Hyperparameters m and r: The maximum intrinsic rank of PERT-RaMa equals $m \cdot r$. We evaluate PERT-RaMa at different intrinsic ranks in $\{4, 8, 16, 32, 64, 128\}$, where m and r have various combinations of values, as shown in Fig. 5. When one of m and r remains unchanged and the other gradually increases, mR and training parameters are basically on an increasing trend. Trainable parameters scale linearly with r and exhibits a cubic relationship with m, which is consistent with the parameter analysis in Table II. Under a fixed intrinsic rank, different combinations of m and r have a significant impact on mR and parameters. For example, at the intrinsic rank of 128, mR only decreases by 0.5, while the number of parameters is reduced by 94% when (m, r) changes from (128, 1) to (8, 16). Considering the retrieval performance and the number of parameters, the optimal combinations of m and r for each intrinsic rank are reported in Table VII. Therefore, how to determine suitable values of m and r to construct the subspace with an appropriate intrinsic rank for RSITR remains a worthwhile consideration.

2) Analysis on Training Method: Fig. 6 shows training time consumption and retrieval performance for the discussed methods in traditional training method BiMa and proposed training method RaMa. Obviously, RaMa decreases training time for all PERT modules to less than (1/5) of that of BiMa. The training time consumption varies from over 200 s to less than 40 s, making models easier to iterate and more resource-friendly. Due to slower calculation speed and higher computational complexity of Kronecker product, KPLoRA and KronA naturally require more training time, while RaMa helps narrow this time gap. Meanwhile, RaMa models RSITR as RaMa, which is more consistent with current evaluation benchmarks, thereby enhancing retrieval recall. RaMa only sacrifices some acceptable memory usage to obtain significant improvements in training speed and bonus gains in retrieval recall. Moreover, it should be mentioned that training time may vary slightly due to hardware instability.

3) Analysis on Matrix Initialization: The performance of LoRA-style fine-tuning methods is always sensitive to the choice of matrix initialization [26], [27]. Therefore, LoRA and KPLoRA with different ways of matrix initialization are evaluated and reported in Fig. 7. Here, we discuss the matrix initialization for matrix A of LoRA and matrices A and α of

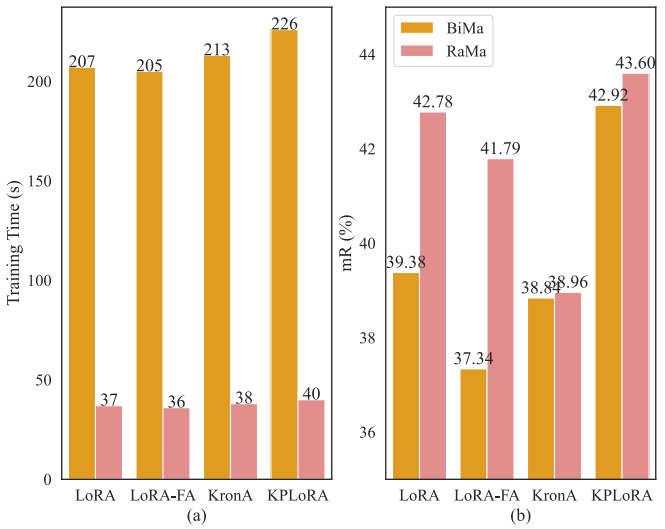


Fig. 6. Results of training time and mR with different training methods on RSITMD dataset. (a) Significant reduction in average training time consumption over 30 epochs of PERT modules. (b) Variation in mR of PERT modules.

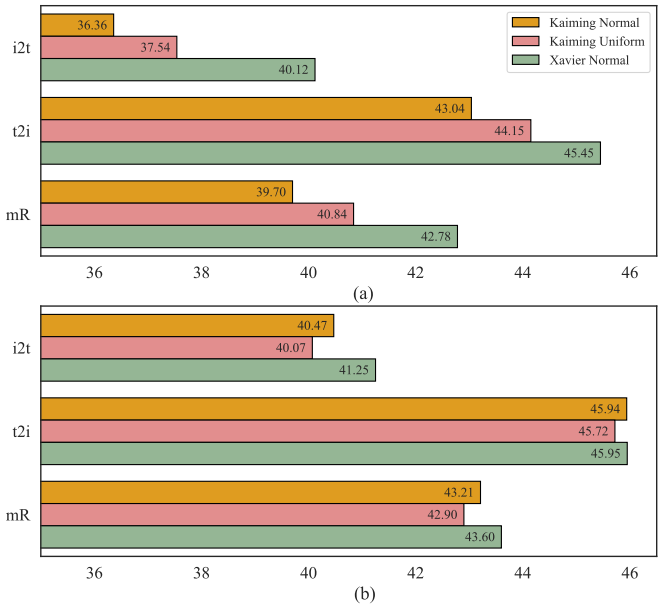


Fig. 7. Results of (a) LoRA and (b) our KPLoRA with various matrix initializations on the RSITMD dataset.

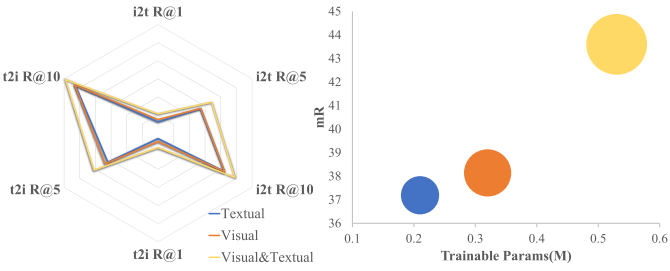


Fig. 8. Results of different encoders with KPLoRA on RSITMD dataset. "Textual" and "Visual" represent the insertion of KPLoRA into only one of the respective encoders of CLIP. (Left) Quantitative metrics $R@K$ of two sub-tasks. (Right) Retrieval performance mR versus the number of trainable parameters.

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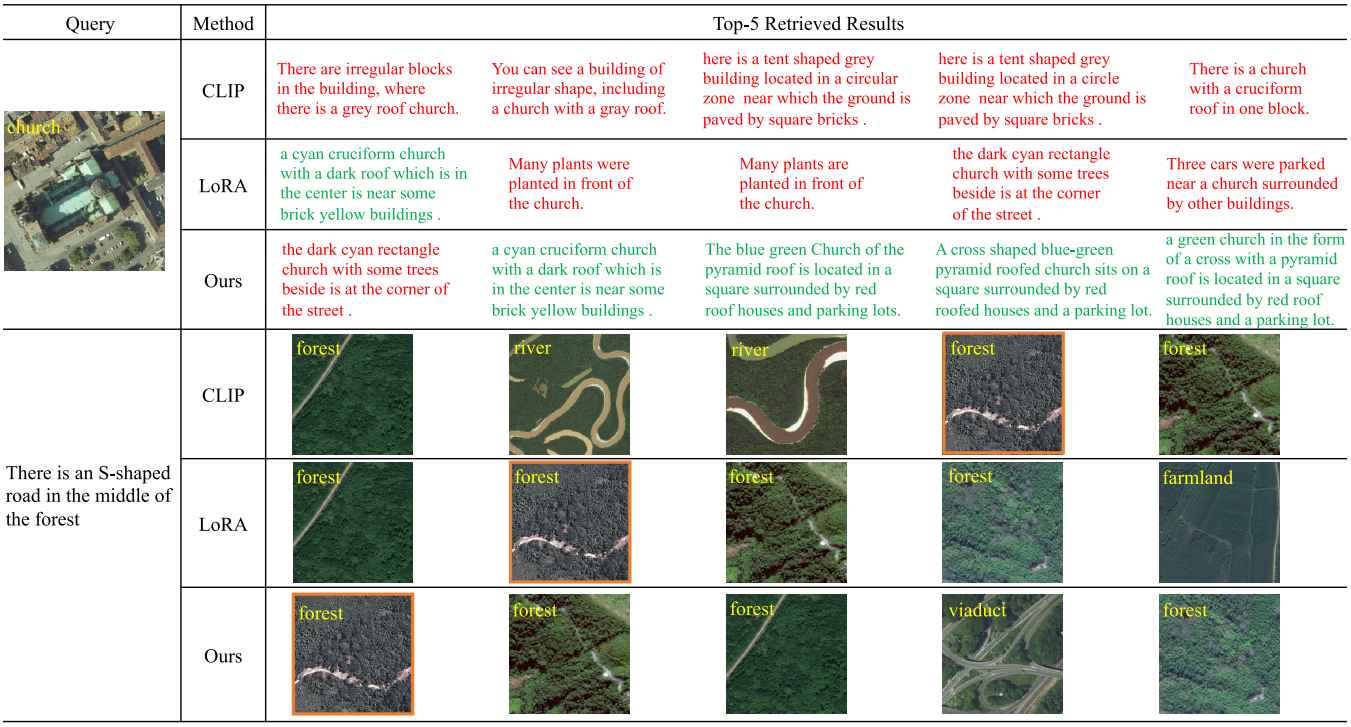


Fig. 9. Visualization of Top-5 *i2t* and *t2i* retrieved results on the RSITMD dataset. The positive text captions of the image query are highlighted in green and the negative in red. The ground-truth image of the text query is marked with the orange box. Words in yellow are the scenario category of RS images provided by the RSITMD dataset.

TABLE VIII

ANALYSIS ON CROSS-LAYER SHARING MECHANISM OF THE PROPOSED KPLORA ON RSITMD DATASET

Share A	Share α	Params	i2t mR	t2i mR	mR
×	×	0.9834M	39.00	45.81	42.40
✓	×	0.9831M	39.40	45.42	42.41
×	✓	0.5329M	39.96	45.91	42.93
✓	✓	0.5325M	41.58	45.95	43.60

KPLoRA. Matrix B of LoRA and β of KPLoRA are both initialized with zeros, maintaining the original output at the beginning of training and converging better. As shown in Fig. 7, the Xavier Normal distribution achieves better retrieval performance on RSITMD dataset compared with the Kaiming Normal distribution and the Kaiming Uniform distribution. It indicates that the Xavier Normal distribution is more suitable for adapting CLIP with LoRA and KPLoRA to RSITR task. Therefore, the Xavier Normal distribution is set as default for matrix initialization in the article.

4) Analysis on Cross-Layer Sharing Mechanism: Furthermore, we conduct an analysis on the impact of the cross-layer sharing mechanism, as shown in Table VIII. Sharing A across layers has a minor impact on mR but contributes to balancing the performance of *i2t* and *t2i*. Sharing α across layers reduces trainable parameters by half and obtains nearly 1% improvement in *i2t* mR. Simultaneous cross-layer sharing A and α can incorporate the above advantages and achieve better recall. A conclusion can be drawn that cross-layer sharing A and α not only realize the complementarity of cross-layer shared knowledge and layer-specific information but also boost retrieval performance with roughly half the trainable parameters.

5) Analysis on Inserted Encoders: Then, we investigate the performance differences of inserting KPLoRA into a single encoder versus dual encoders of CLIP, as shown in Fig. 8. Intuitively, inserting KPLoRA solely in the textual encoder results in worse performance compared to inserting it only in the visual encoder. It is due to RS images exhibiting a greater domain and distribution discrepancy from pretraining images of CLIP compared to text captions. Although there is no direct cross-modal interaction, inserting KPLoRA into both visual and textual encoders clearly improves the ability to capture the complex correspondence between RS images and text captions.

E. Visualization

To intuitively demonstrate and qualitatively compare the effectiveness of the proposed method, we present a series of RSITR examples in Fig. 9. Zero-shot CLIP and LoRA are selected as the comparison methods. As shown in Fig. 9, zero-shot CLIP completely fails to retrieve relevant text captions for this image query, while LoRA only searches one matched text caption at Top-1, and our method achieves the highest recall within Top-5 results. Three methods can basically understand the main object "church" in the RS image and associate it with relative texts. However, zero-shot CLIP and LoRA make mistakes in correlating partial regions of the RS image with the partial caption. After PEFT with our PERT-RaMa, the CLIP-based retrieval model acquires detailed information from complex backgrounds and bridges the semantic gap, recalling the most relevant text captions.

As for the text query *"There is an S-shaped road in the middle of the forest,"* the ranking of the ground-truth

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image in the retrieved sequence obviously illustrates our method is better at establishing a match between text and image. Zero-shot CLIP confuses "river" and "road" in RS scenarios. LoRA effectively distinguishes these two concepts, but still mis-classifies "forest" and "farmland" due to the similar appearance. Top-5 retrieved RS images of our method typically include "S-shaped road" and "forest," although the forth belongs to the scenario category of viaduct in RSITMD dataset. Therefore, our proposed method better retrieves RS images containing all attributes of the text query, alleviating the domain gap when transferring CLIP into RS domain.

In summary, our proposed method exhibits effective and robust capabilities in ranking for image–text retrieval and matching for text–image retrieval.

V. Conclusion and Future Work

In this article, we propose a novel framework, PERT-RaMa, exploring an alternative PEFT paradigm for RSITR task. The integration of our proposed KPLoRA into the Transformer blocks of CLIP does not modify the inherent architecture and add any inference latency by merging KPLoRA into original weights. KPLoRA is built on Kronecker products and acquires a higher intrinsic rank with fewer parameters to address the overfitting issue and acquire superior performance. RaMa considers the RSITR task as an asymmetric RaMa problem and optimizes the model with a corresponding loss function, which removes redundant computation and accelerates model training.

However, the intrinsic rank and the combination of hyperparameters depend on manual setting for KPLoRA. The choices of insertion location and intrinsic rank are based on prior experience. How to design a more flexible and dynamic PERT module remains to be explored. And the proposed method fully utilizes the characteristic of RSITR datasets but ignores the inherent attributes of RS images, e.g., multiscale, unbalanced distribution, and low contrast of foreground and complex background. In the future, our focus will be directly toward designing more flexible, scalable, and adaptable PEFT methods. We will extend the research into various cross-modal vision-language tasks in RS domain and further explore practical applications with the combination of RS images and natural languages. Moreover, the emergence of satellite videos offers a feasible approach for continuous Earth observation. How to implement satellite video text retrieval to establish the correlation between temporal dynamic information and natural languages will be a key direction for our future research.

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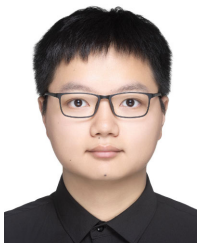
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